FortyGuard Heat Intelligence Layer for QGIS

Project report: a QGIS plugin that brings FortyGuard hyper-local temperature intelligence to any map project

Plugin version	0.4.0 (0.3.3 is the version currently approved and public on plugins.qgis.org; newer versions are uploaded and awaiting approval)
Status	Published in the official QGIS Plugin Repository (experimental), 15 October 2026
Source code	https://github.com/FortyGuard-Tech/qgis-fortyguard-heatmap (GPL-2.0-or-later)
Listing	https://plugins.qgis.org/plugins/fortyguard_heatmap/
Tested on	macOS 27 (Apple Silicon), QGIS 3.44.15 LTR
Prepared by	Mustafa Hamdy, Automation Engineer, FortyGuard - 16 October 2026

1. Summary

FortyGuard is a temperature intelligence layer for the physical world: it estimates the local thermal state at the level of a site, an asset or a route (a building, a data center, a city block) instead of the nearest weather station, so temperature can inform physical decisions and Physical AI systems. One intelligence foundation powers four products: Temperature Dashboard (visual exploration for professionals and teams), Temperature API (modeled temperature, forecasts and thermal analytics for software, data and geospatial products), Temperature MCP (permissioned access for AI agents through the Model Context Protocol) and Temperature Enterprise (recurring, governed thermal intelligence across assets, portfolios and organizations).

FortyGuard Heat Intelligence Layer is a QGIS plugin that brings the Temperature API into QGIS: any QGIS user can run FortyGuard heat analyses on a polygon in their own map project, using their own FortyGuard API key. The user selects an area, picks an analysis, a time window and a resolution, clicks Generate Heatmap, and receives a styled temperature layer in the same project. The plugin talks directly to the FortyGuard Temperature API (POST /v1/heatmap, GET /v1/status/{activity_id}); no intermediate backend is involved.

Example, a data center site: draw the site boundary with the surrounding blocks. A single-day Temperature layer shows which side of the site sits in the hottest tiles; Exceedance with a 35 °C threshold over a summer month counts how many hours the surroundings spend above that limit; Persistence shows the longest continuous hot spell. Together they help evaluate cooling load, capacity margins and operating constraints before a physical decision is made.

What was delivered:

- A complete, installable QGIS plugin with four analysis modes (Temperature/Snapshot, Exceedance, Persistence, Peak temperature time).
- 131 automated tests plus automated checks inside the real QGIS application.
- Live verification against the FortyGuard API (development environment): five completed requests covering all four analyses.
- A public source repository with versioned releases, an installation guide, an API contract document and a test report.
- Submission to the official QGIS Plugin Repository: security scan passed, email confirmed, approved and publicly listed.

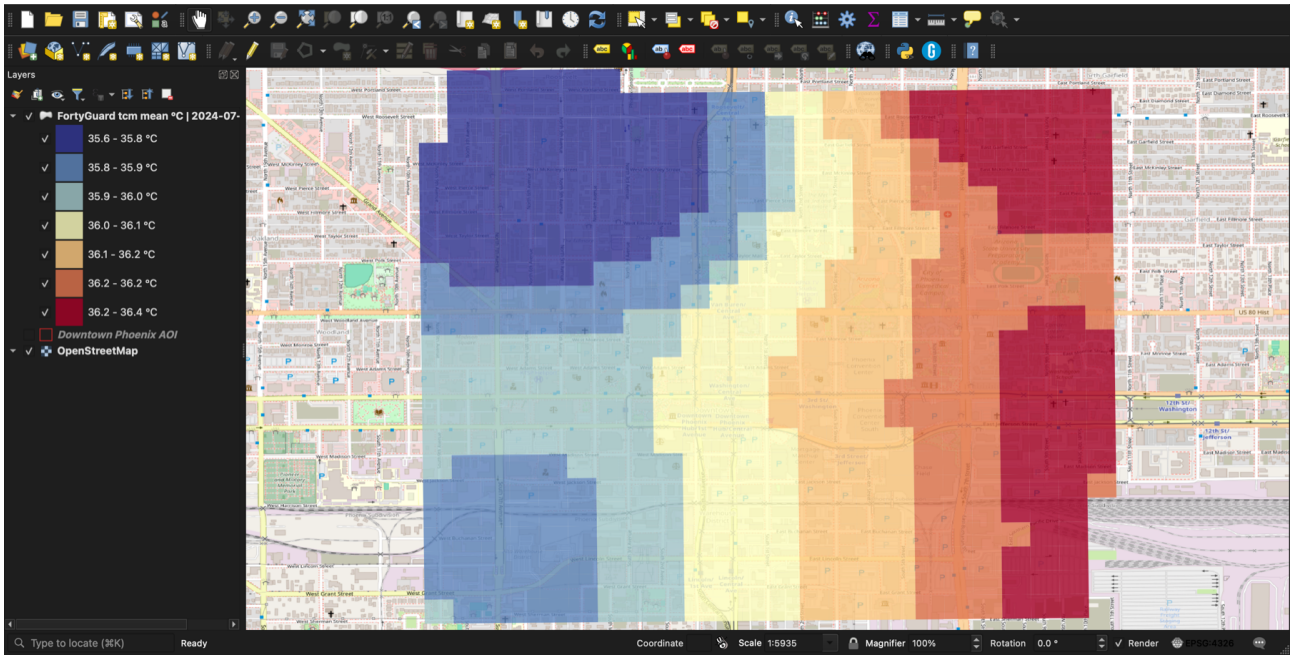


Figure 1. Real FortyGuard temperature heatmap (tcm, daily mean, 100 m tiles) over downtown Phoenix, Arizona, 15 July 2024, inside QGIS 3.44: the plugin's layer with its °C legend in the Layers panel, on top of OpenStreetMap.

2. How users install the plugin

The plugin is distributed through the official QGIS Plugin Repository, so installation is the standard QGIS flow:

- Open QGIS and choose **Plugins > Manage and Install Plugins...**
- In **Settings**, tick **Show also experimental plugins** (required while the release is marked experimental).
- In the **All** tab, search for **FortyGuard**, select **FortyGuard Heat Intelligence Layer** and click **Install Plugin**.
- A FortyGuard button appears in the toolbar and under **Plugins > FortyGuard Heat Intelligence Layer**.
- Offline alternative: **Install from ZIP** with the release file from GitHub (fortyguard_heatmap-<version>.zip).

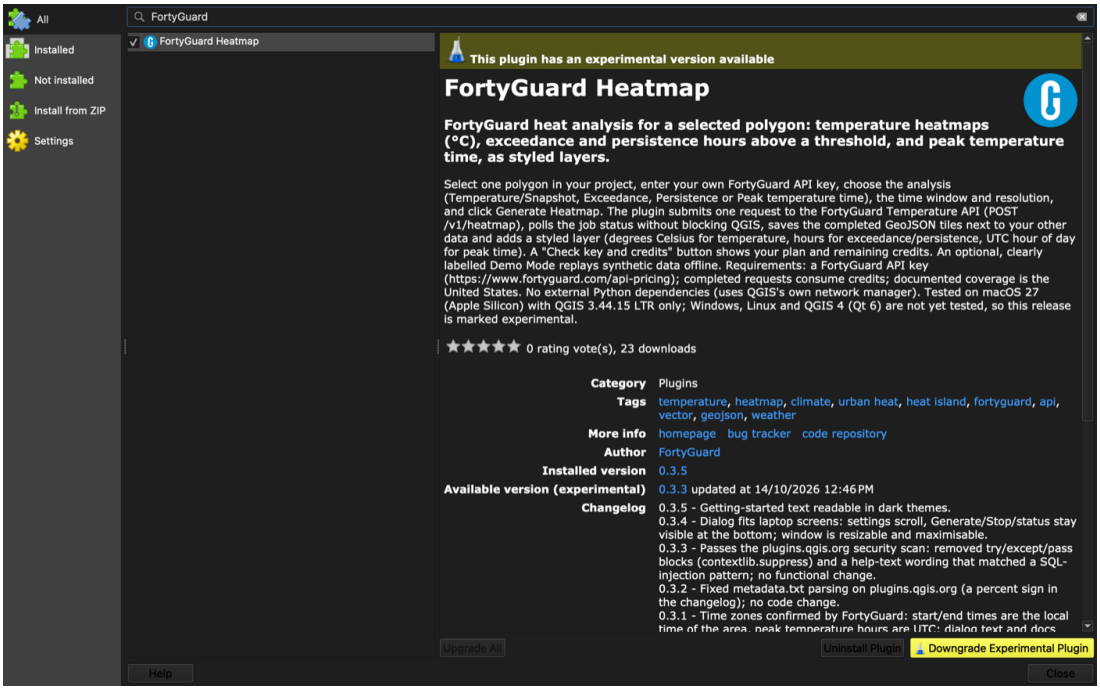


Figure 2. The QGIS Plugin Manager with FortyGuard Heat Intelligence Layer selected (listed as FortyGuard Heatmap until the renamed version is approved): description, tags, links, versions and the install button.

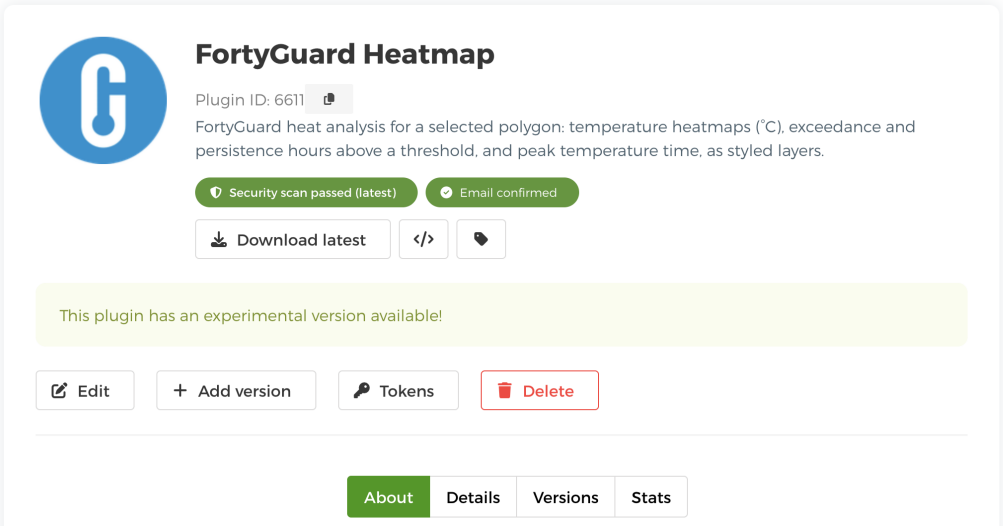


Figure 3. The plugin page on plugins.qgis.org after approval: security scan passed, email confirmed, experimental version available.

3. How a heatmap is generated

Everything happens in one dialog. The settings scroll; the Generate button, status and log stay visible at the bottom.

- **API key**: the user pastes their own key (masked). **Check key and credits** shows the plan, key status, expiry and remaining credits before anything is spent. The key stays in memory; optionally it can be saved in the encrypted QGIS authentication database.
- **Area of interest**: any polygon layer in the project; exactly one selected feature. The plugin transforms it to EPSG:4326 (holes preserved) and shows the source CRS and area.
- **Analysis**: Temperature/Snapshot, Exceedance, Persistence or Peak temperature time, with a one-line explanation. A threshold (°C, default 35) appears only for Exceedance and Persistence.
- **Date and time**: single hour, range of hours, single day, range of days or calendar month, validated against the documented limits (2019-01-01 to now + 12 h, 23 h, 31 days). Times are the local time of the area.
- **Output**: 60 / 80 / 100 m resolution, measure for daily outputs (mean/min/max), output folder, maximum wait.
- **Generate Heatmap**: a confirmation shows the environment, area, request and credit cost. The job is polled in the background (QGIS stays responsive); Stop Waiting stops polling only, Check Again resumes the same activity.
- **Result**: GeoJSON saved (never overwritten) with a metadata sidecar, loaded as a separately styled layer, statistics shown in the dialog.

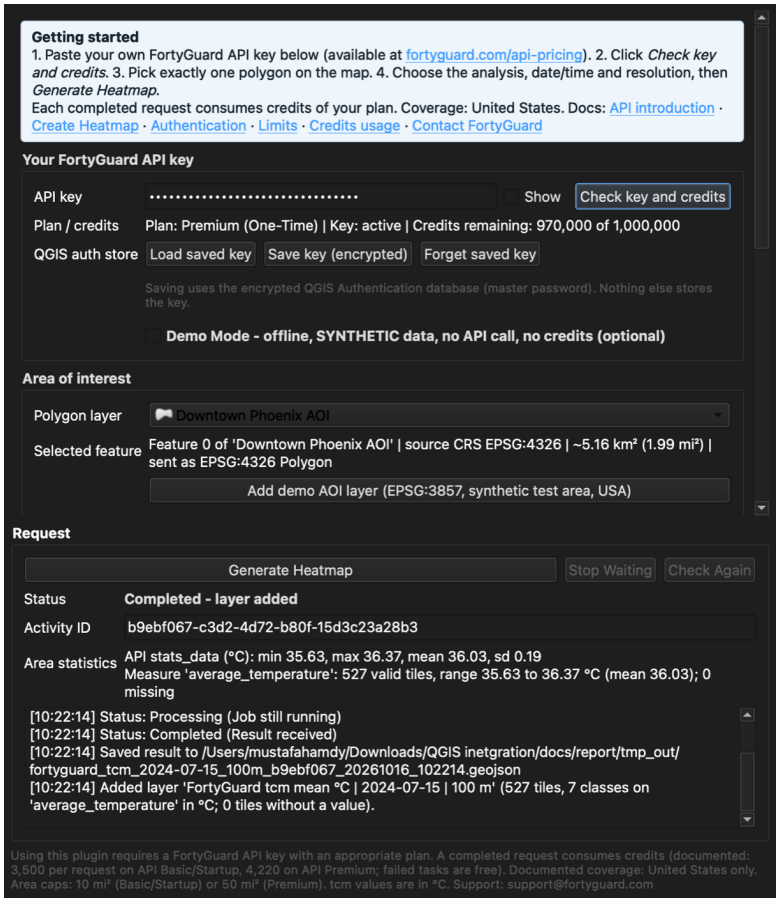


Figure 4. The plugin dialog after a completed request for downtown Phoenix: 527 tiles, daily mean 35.63 to 36.37 °C, activity ID and saved file shown in the log.

4. Result on the map

The test area is a 5.16 km² (1.99 mi²) rectangle over downtown Phoenix, Arizona. Figure 5 shows the base map with the area outline; Figure 6 shows the heatmap the plugin produced for 15 July 2024 at 100 m resolution, with the selection layer hidden.

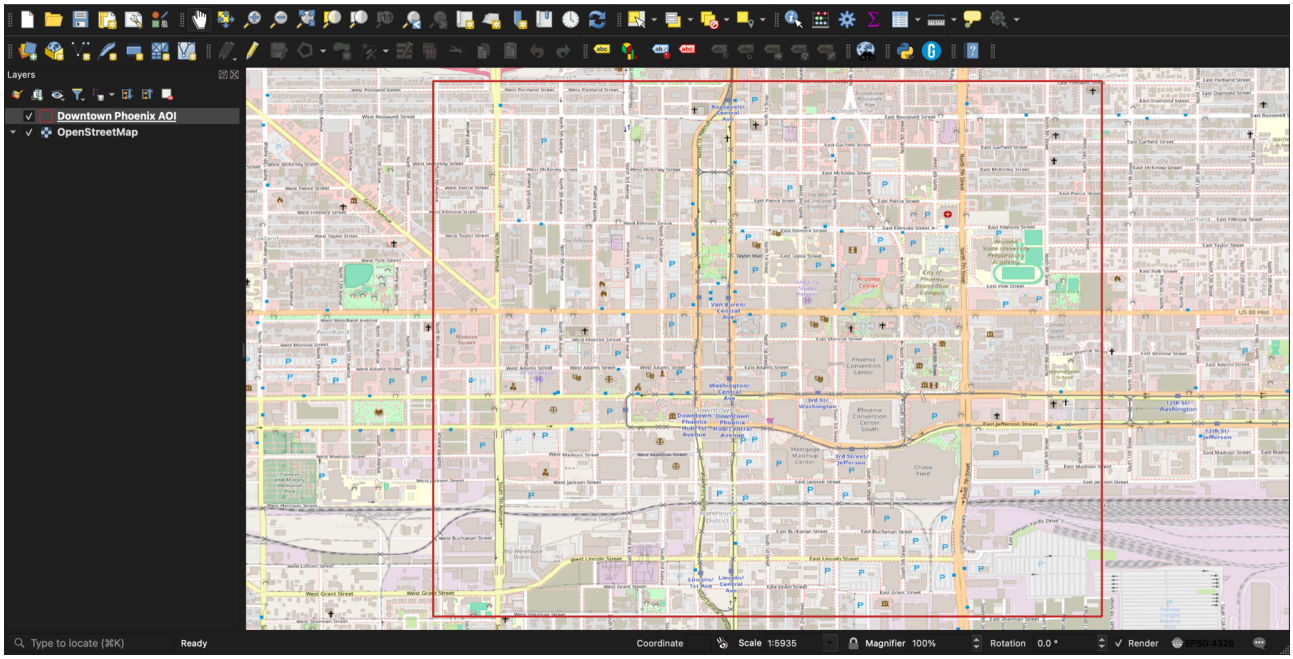


Figure 5. QGIS with the OpenStreetMap base map of downtown Phoenix and the selected area outlined in red.

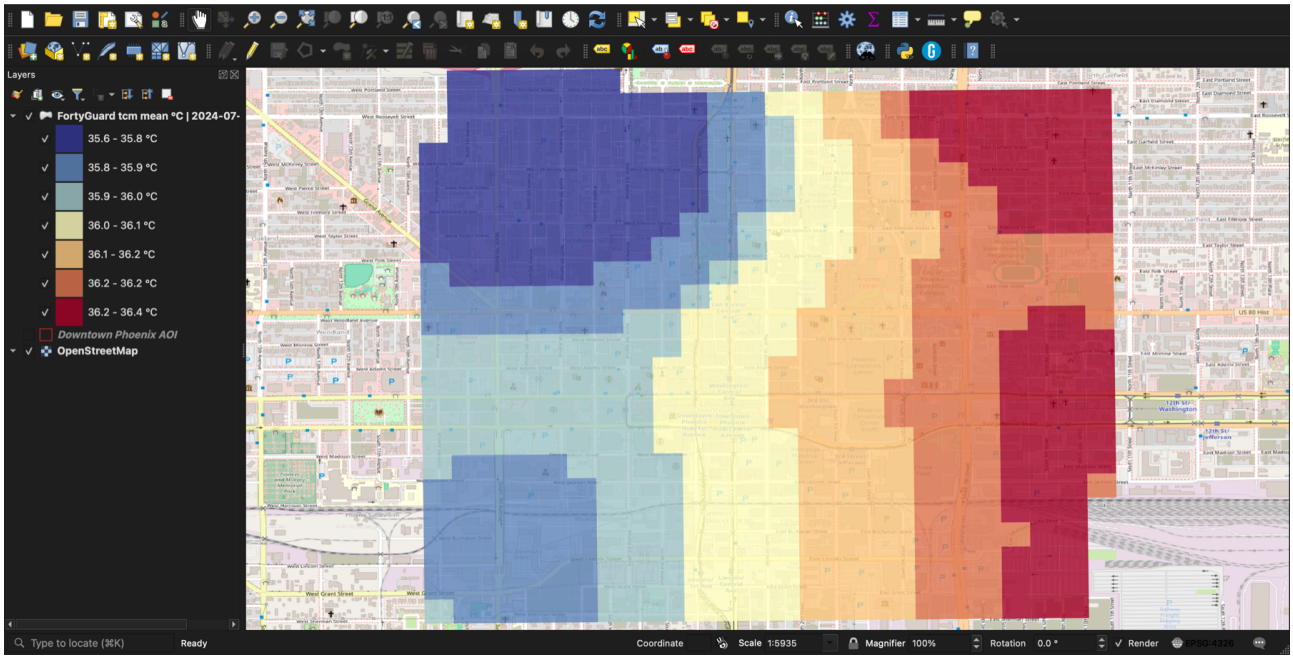


Figure 6. The same QGIS project after Generate Heatmap: the FortyGuard tcm layer (daily mean, °C) with its seven-class legend (35.6 to 36.4 °C) in the Layers panel. Blue tiles are cooler, red tiles are hotter. Data returned by the FortyGuard API, activity b9ebf067.

5. Analysis modes

The four modes mirror the FortyGuard Autodesk Forma extension. Each mode is one API request; the API computes the result.

Mode	analytic_type	Result per tile	Legend
Temperature / Snapshot	tcm	temperature in the period (°C); mean / min / max for daily windows	°C
Exceedance	exceedance	total hours above the threshold in the period	hours
Persistence	persistence	longest continuous run of hours above the threshold	hours
Peak temperature time	time_of_measure	hour of day (0-23) when the tile reaches its peak	clock classes, UTC

- Threshold and direction ("above") are sent only for Exceedance and Persistence.
- Single hour is disabled for the three analyses; the dialog switches to Single day and says so.
- Calendar month submits the month's first and last day as a range of days (leap years handled, current month clipped).
- Zero hours and 00:00 are valid results, not missing data. Percentages in the summary are shares of valid tiles, not of area.
- Each result is its own layer, named with the analysis, dates, resolution and threshold.
- Appearance: Smooth surface (default, dashboard-like continuous raster, tiles kept hidden) or Tiles only.

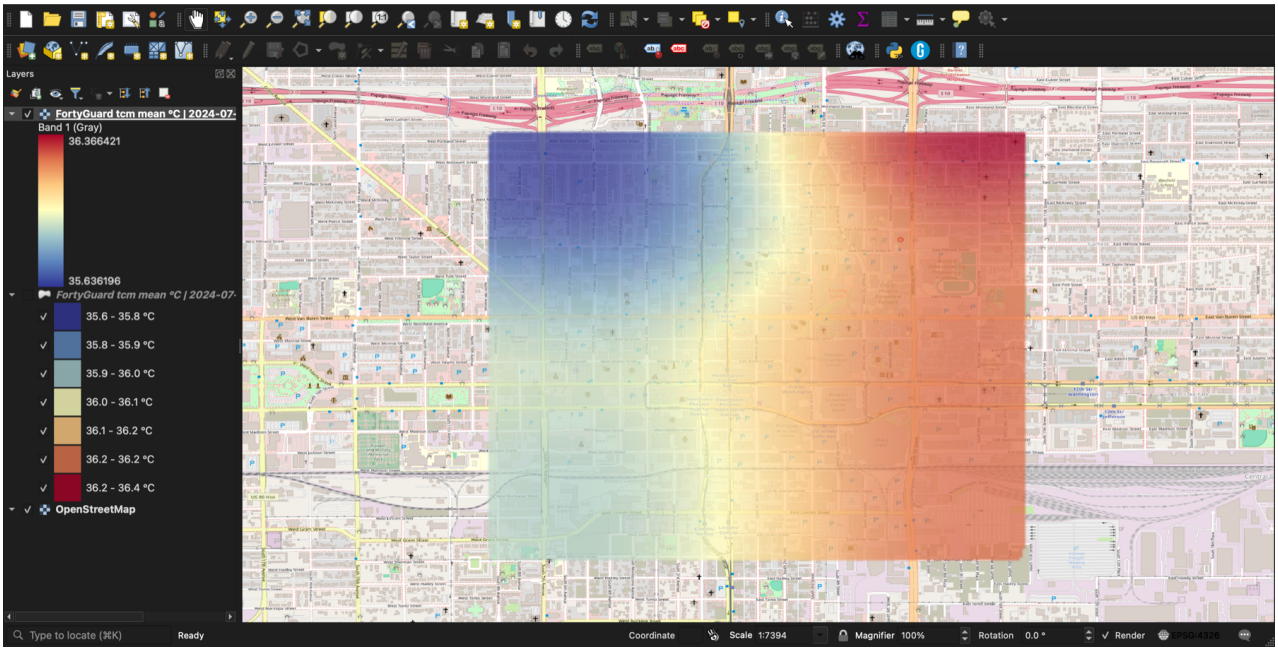


Figure 7a. Smooth surface appearance (0.4.0): the same 527 Phoenix tiles interpolated into a continuous raster, clipped to the area, with the exact tiles layer kept hidden underneath. Presentation only; values between tile centres are estimates.

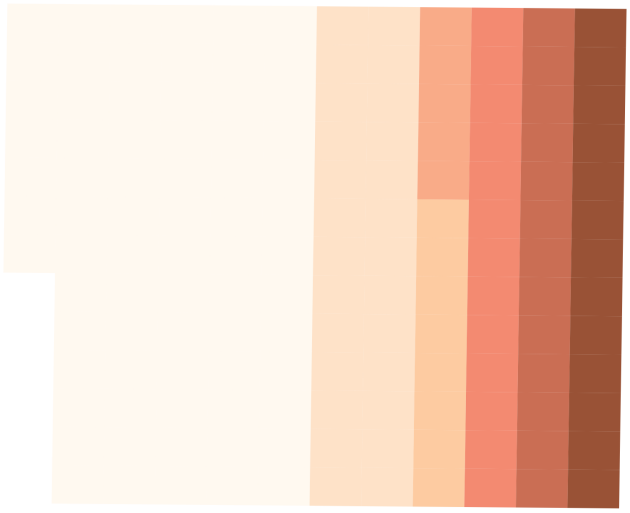


Figure 7b. Live Exceedance result (hours above 35 °C, 15-21 July 2024, lower Manhattan test area) rendered with the plugin's hours legend. Returned by the FortyGuard API, activity d0439dfd.

6. Technical design

- Pure QGIS/Python plugin, no external dependencies; network calls use QgsNetworkAccessManager (asynchronous, cache disabled for API calls, redirects not followed).
- Modules: contract (verified API constants), payload (request building and validation), responses (envelope and schema parsing), polling (bounded deadline, backoff, retry limits), geometry (selection and CRS transformation), api_client, job (submit once, poll with timers), output (save and load), styling (graduated renderers), credentials, demo.
- The API key is sent only as the documented api-key header to the chosen FortyGuard host and never written to logs, layers, project files or result files. Production is the default; the development environment sits in a separate Developer settings section and switching clears the key.
- A submission is never repeated automatically; the activity ID is kept so an interrupted job can be checked again.
- Result schemas are validated per response (temperature fields for the snapshot, the value field for analyses) instead of being assumed.
- Demo Mode replays clearly labelled synthetic fixtures through the same code path for offline testing.

7. Validation

Check	Result
Automated tests (pure Python + QGIS runtime)	131 tests, 130 passed, 1 opt-in skipped
Automated checks inside QGIS 3.44.15	Install/load, Demo Mode, styling, project save/reopen, unload, Plugin Manager install: passed
Live API, development environment	tcm single day and single hour; exceedance, persistence and time_of_measure (range of days): all completed; schemas and units verified
Real request through the dialog	Completed (tcm, lower Manhattan); later Phoenix runs by the user
plugins.qgis.org security scan	Passed (Bandit and secret detection clean)
Not yet tested	Windows, Linux, QGIS 4 (Qt 6), production environment (development key only)

8. Publication timeline

Date	Milestone
12 Oct 2026	API documentation verified; prototype with live request path and Demo Mode
13 Oct 2026	First live requests (dev); release candidate 0.2.0; public GitHub repository created; analysis modes 0.3.0; time zones confirmed by the API team (0.3.1)
14 Oct 2026	OSGeo account; 0.3.2 uploaded and blocked by the security scan; 0.3.3 uploaded and passed; plugin email confirmed
15 Oct 2026	0.3.3 approved and publicly listed; Plugin Manager install verified; 0.3.4/0.3.5 (laptop-screen layout, dark-theme text) released on GitHub and uploaded

9. API findings reported to the FortyGuard team

- An hourly tcm request returned average/min/max fields (identical values) instead of the documented temperature field.
- Exceedance and persistence values are fractional hours.
- Error messages are under details.message, not the top-level message field.
- The usage endpoint returns HTTP 500 on production for a development key.
- Confirmed by the team: start/end times are the local time of the area; peak-time hours are UTC. The public docs do not yet state the input time zone.

10. Open items and next steps

- Approval of the latest upload (0.4.0: FortyGuard Heat Intelligence Layer, new listing text, smooth surface) on plugins.qgis.org. Users on 0.3.3 will see an Upgrade button.
- Windows and Linux test runs, then a release without the experimental flag so the plugin appears in the default Plugin Manager search.
- One end-to-end request on the production API with a production key.
- Optional: 'below' threshold direction, local-time display of peak hours, other FortyGuard products (satellite, street view, heat intelligence).

11. Links

- Plugin page: https://plugins.qgis.org/plugins/fortyguard_heatmap/
- API documentation: https://docs-api.fortyguard.com/docs/introduction (Create Heatmap: https://docs-api.fortyguard.com/docs/create-heatmap)
- Source and releases: https://github.com/FortyGuard-Tech/qgis-fortyguard-heatmap
- FortyGuard API documentation: https://docs-api.fortyguard.com/docs/introduction
- API plans: https://www.fortyguard.com/api-pricing - Support: support@fortyguard.com